Note 4.3 Monotonic Functions and the First Derivative Test

Increasing and Decreasing Functions

- **Definition:**
  Suppose that f is continuous on $[a, b]$ and differentiable on (a, b), if $f'(x) > 0(f'(x) < 0)$ at each point $x \in (a, b)$, then f is increasing (decreasing) on $[a, b]$.
- **The Procedure of Finding Monotone Intervals**
  - Find the point where $f'(c) = 0$ or $f'(x)$ doesn't exist.(critical points)
  - Make a sign table
  - From the table determine the monotone interval and relative extrema directly.

First Derivative Test for Local Extrema

- If f' changes from negative(positive) to positive(negative) at c, then f' has a local minimum(maximum) at c.
- If f' doesn't change sign at c, no local extremum at c.